[Total: 8]

2 **(a)** State the conditions required for a body to be in equilibrium.

When a body is in equilibrium, the following two conditions must both be satisfied:

1. The resultant force on it must be zero. [B1]

2. The resultant torque on it must be zero. [B1]

[2]

(b) Fig. 2.1 shows a lamp weighing 5.00 N that is hung from the end of a beam 4.50 m long and weighing 1.00 N, making an angle of 25.0° below the horizontal.

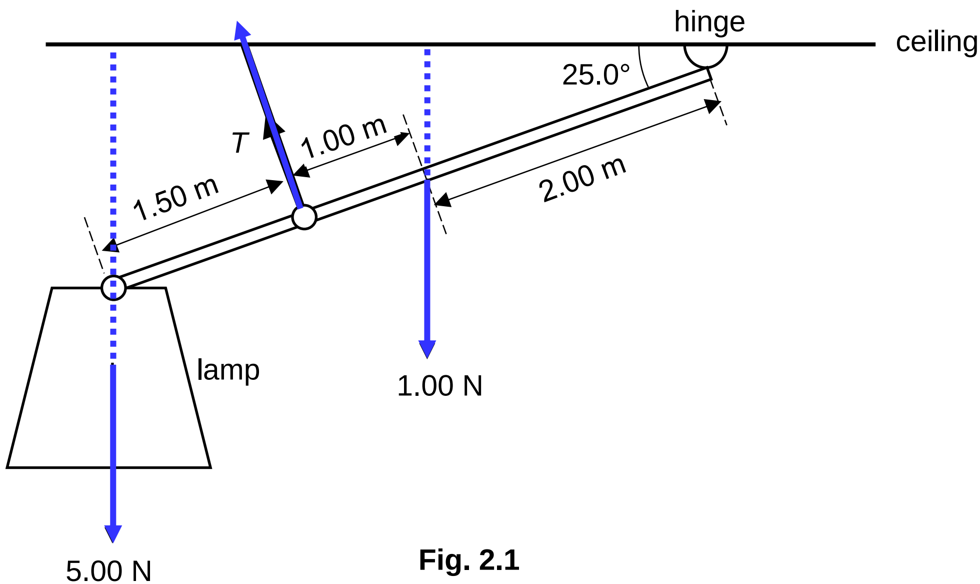

Fig. 2.1

The beam is held in position by a hinge at its upper end and by a cable 3.00 m lower down the beam and perpendicular to it. The centre of gravity of the beam is 2.00 m down the beam from the hinge.

(i) The position of the centre of gravity of the beam is not at its midpoint. Suggest what this implies about the distribution of the mass in the beam.

The distribution of mass in the beam would be non-uniform. [B1]

....

....

.... [1]

(ii) Show that the tension *T* in the cable is 7.40 N. [2]

taking moments about the pivot

1.00 (2.00 cos 25.0) + 5.00 (4.5 cos 25.0) -T(3.00) = 0 [B1]

T = 7.40 N [A1]

(iii) Determine the magnitude and the direction of the force acting on the beam at the hinge.

taking moments about the pivot

1.00 (2.00 cos 25.0) + 5.00 (4.5 cos 25.0) -T(3.00) = 0

where the forces are all in kN

T = 7.40 kN

(the minus sign meens that this force is actually acting downwards)

Assume the horizontal component of the force at the hinge is F_x to the right

$F_x + 7.40 \sin 25.0° = 0$

$F_x = 3.13$ N [B1 for finding F_X and F_Y]

$$F = \sqrt{F_x{}^2 + F_Y{}^2}$$

$$= \sqrt{3.13^2 + 0.707^2}$$

$$= 3.21 \text{ N}$$ [B1 for finding F]

Let θ be the angle of F <u>below the horizontal</u>

$$\tan \theta = \frac{0.707}{3.10}$$

$\theta = 12.8°$ [B1 for finding θ and describing direction]

magnitude = ...

direction = ... [3]

[Total: 8]

3 The Earth may be assumed to be a uniform sphere of radius R and mass M. At its surface, the